

February 16, 2026

Important for partial derivatives: In practice $\frac{\partial f}{\partial x_i}$ is found by

- 1) Treat all variables other than x_i as constants
- 2) Take derivative as in Calc I using derivative rules
- 3) If all points in the domain of f are in the domain of $\frac{\partial f}{\partial x_i}$, then you are done
- 4) For any point $\vec{a}$ in the domain of f but not in the domain of $\frac{\partial f}{\partial x_i}$ you found, you may need to apply the partial derivative definition

$$\text{e.g. } f(x, y) = \begin{cases} 0 & , \text{ if } (x, y) = (0, 0) \\ \frac{xy}{x^2 + y^2} & , \text{ if } (x, y) \neq (0, 0) \end{cases}$$

For $(x, y) \neq (0, 0)$

$$\frac{\partial f}{\partial x} = \frac{y x^2 + y y^2 - x y 2x}{(x^2 + y^2)^2} = \frac{y^3 - x^2 y}{(x^2 + y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{x(x^2 + y^2) - x y 2y}{(x^2 + y^2)^2} = \frac{x^3 - x y^2}{(x^2 + y^2)^2}$$

For $(x, y) = (0, 0)$

$$\frac{\partial f}{\partial x}(0, 0) = 0 = \frac{\partial f}{\partial y}(0, 0)$$

using the definition of part. derivative

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{\frac{h(0)}{h^2 + 0^2} - 0}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, 0+h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

The Gradient

def: The gradient of a function $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is the vector of partial derivatives
 $\nabla f(\vec{x}) := (\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n})$ (sometimes denoted $\vec{\nabla} f$)

e.g.: $f(x, y, z) = 2xy^2z$

$$\frac{\partial f}{\partial x} = 2y^2z$$

$$\frac{\partial f}{\partial y} = 2xz \cdot 2y = 4xyz$$

$$\frac{\partial f}{\partial z} = 2xy^2$$

$$\vec{\nabla} f(x, y, z) = (2y^2z, 4xyz, 2xy^2)$$

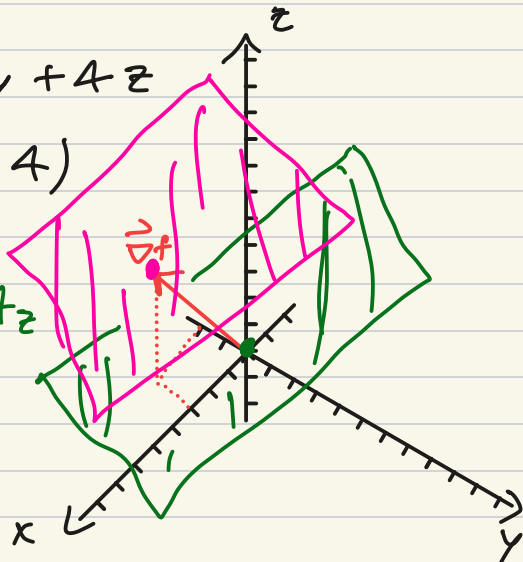
e.g.: $f(x, y, z) = 3x - 2y + 4z$

$$\vec{\nabla} f(x, y, z) = (3, -2, 4)$$

level sets $c = 0 = 3x - 2y + 4z$

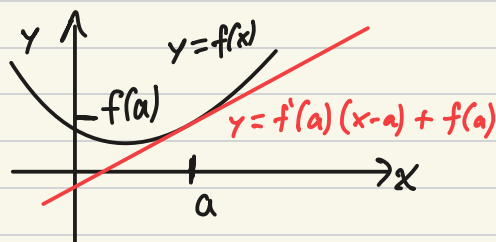
$$c = 21 = 3x - 2y + 4z$$

the function f increases the most in the direction of $\vec{\nabla} f$



Differentiability

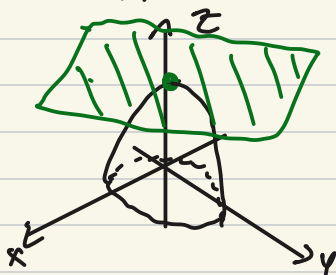
The derivative of f at a allows us to approximate f around a by its tangent line



Tangent plane to a function $f: \mathbb{X} \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$

intuitively: f is differentiable at $\vec{a}$ if we can approximate f around $\vec{a}$ by a linear function (i.e. its tangent plane)

e.g.: $f(x,y) = 4 - x^2 - y^2$ at $\vec{a} = (0,0)$



$$\vec{\nabla} f(x,y) = (-2x, -2y)$$

$$\vec{\nabla} f(0,0) = (0,0)$$

$$\text{plane: } (0,0,1) \cdot (x,y,z) = f(0,0) \\ z = 4$$

def: A scalar function $f: \mathbb{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a} \in \mathbb{X}$ if there is a linear function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ (i.e. $h(x_1, \dots, x_n) = c + a_1 x_1 + a_2 x_2 + \dots + a_n x_n$) such that
$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x}) - h(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0$$

Theorem: If $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$ then the best linear approximation to f around $\vec{a}$ (i.e. $h(\vec{x})$) is $h(\vec{x}) = \vec{\nabla} f(\vec{a}) \cdot (\vec{x} - \vec{a}) + f(\vec{a})$

e.g. : Find the linear approximation of $f(x, y) = 4x^2y + y^2$ at $(x, y) = (1, 2) = \vec{a}$
 $h(\vec{x}) = ?$

$$\vec{\nabla} f = (8xy, 4x^2 + 2y)$$

$$\begin{aligned}\vec{\nabla} f(\vec{a}) &= (8(1)(2), 4(1)^2 + 2(2)) \\ &= (16, 8)\end{aligned}$$

$$f(\vec{a}) = 4(1)^2(2) + (2)^2 = 12$$

$$\begin{aligned}h(\vec{x}) &= h(x, y) = (16, 8) \cdot (x - 1, y - 2) + 12 \\ &= 16x - 16 + 8y - 16 + 12 \\ &= 16x + 8y - 20\end{aligned}$$